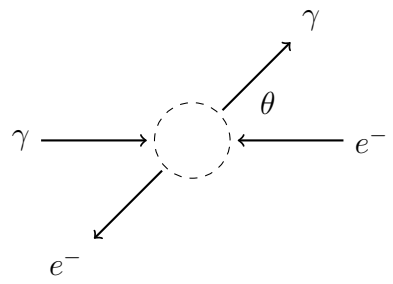
Compton scattering 1

Compton scattering is the process $\gamma + e^- \to \gamma + e^-$.



Momentum vectors where $E = \sqrt{\omega^2 + m^2}$.

$$p_1 = \underset{\gamma \longrightarrow}{\begin{pmatrix} \omega \\ 0 \\ 0 \\ \omega \end{pmatrix}} \qquad p_2 = \underset{\longleftarrow e^-}{\begin{pmatrix} E \\ 0 \\ 0 \\ -\omega \end{pmatrix}} \qquad p_3 = \underset{\nearrow \gamma}{\begin{pmatrix} \omega \\ \omega \sin\theta \cos\phi \\ \omega \sin\theta \sin\phi \\ \omega \cos\theta \end{pmatrix}} \qquad p_4 = \underset{e^- \swarrow}{\begin{pmatrix} E \\ -\omega \sin\theta \cos\phi \\ -\omega \sin\theta \sin\phi \\ -\omega \cos\theta \end{pmatrix}}$$

Spinors for p_2.

$$u_{21} = \frac{1}{\sqrt{E+m}} \underset{\text{spin up}}{\begin{pmatrix} E+m \\ 0 \\ -\omega \\ 0 \end{pmatrix}} \qquad u_{22} = \frac{1}{\sqrt{E+m}} \underset{\text{spin down}}{\begin{pmatrix} 0 \\ E+m \\ 0 \\ \omega \end{pmatrix}}$$

Spinors for p_4.

$$u_{41} = \frac{1}{\sqrt{E+m}} \underset{\text{spin up}}{\begin{pmatrix} E+m \\ 0 \\ p_{4z} \\ p_{4x} + ip_{4y} \end{pmatrix}} \qquad u_{42} = \frac{1}{\sqrt{E+m}} \underset{\text{spin down}}{\begin{pmatrix} 0 \\ E+m \\ p_{4x} - ip_{4y} \\ -p_{4z} \end{pmatrix}}$$

The scattering amplitude $\mathcal{M}_{ab}{}^{\mu\nu}$ for spin ab and polarization $\mu\nu$ is

$$\mathcal{M}_{ab}{}^{\mu\nu} = \mathcal{M}_{1ab}{}^{\mu\nu} + \mathcal{M}_{2ab}{}^{\nu\mu}$$

where

$$\mathcal{M}_{1ab}{}^{\mu\nu} = \frac{\bar{u}_{4b}(-ie\gamma^\mu)(\not{p}_1 + m)(-ie\gamma^\nu)u_{2a}}{s - m^2}$$

and

$$\mathcal{M}_{2ab}{}^{\nu\mu} = \frac{\bar{u}_{4b}(-ie\gamma^\nu)(\not{p}_2 + m)(-ie\gamma^\mu)u_{2a}}{u - m^2}$$

Matrices $\not{q}_1$ and $\not{q}_2$ represent momentum transfer.

$$\not{q}_1 = (p_1 + p_2)^\alpha g_{\alpha\beta}\gamma^\beta$$
$$\not{q}_2 = (p_4 - p_1)^\alpha g_{\alpha\beta}\gamma^\beta$$

Expected density $\langle|\mathcal{M}|^2\rangle$ is the average density.

$$\langle|\mathcal{M}|^2\rangle = \frac{1}{4}\sum_{ab}\sum_{\mu\nu}\left|\mathcal{M}_{ab}{}^{\mu\nu}\right|^2$$

Summing over $\mu\nu$ requires $g_{\mu\nu}$ to lower indices.

$$\langle|\mathcal{M}|^2\rangle = \frac{1}{4}\sum_{ab}\mathcal{M}_{ab}{}^{\mu\nu}\left(g_{\mu\alpha}\mathcal{M}_{ab}{}^{\alpha\beta}g_{\beta\nu}\right)^*$$

Expand the summand.

$$\langle|\mathcal{M}|^2\rangle = \frac{1}{4}\sum_{ab}\Big(\mathcal{M}_{1ab}{}^{\mu\nu}\left(g_{\mu\alpha}\mathcal{M}_{1ab}{}^{\alpha\beta}g_{\beta\nu}\right)^* + \mathcal{M}_{1ab}{}^{\mu\nu}\left(g_{\nu\alpha}\mathcal{M}_{2ab}{}^{\alpha\beta}g_{\beta\mu}\right)^* + \mathcal{M}_{2ab}{}^{\nu\mu}\left(g_{\mu\alpha}\mathcal{M}_{1ab}{}^{\alpha\beta}g_{\beta\nu}\right)^* + \mathcal{M}_{2ab}{}^{\nu\mu}\left(g_{\nu\alpha}\mathcal{M}_{2ab}{}^{\alpha\beta}g_{\beta\mu}\right)^*\Big)$$

The summation results are

$$\mathcal{M}_{1ab}{}^{\mu\nu}\left(g_{\mu\alpha}\mathcal{M}_{1ab}{}^{\alpha\beta}g_{\beta\nu}\right)^* = \frac{e^4}{(s-m^2)^2}\left(-8su + 24m^2s + 8m^2u + 8m^4\right)$$
$$\mathcal{M}_{1ab}{}^{\mu\nu}\left(g_{\nu\alpha}\mathcal{M}_{2ab}{}^{\alpha\beta}g_{\beta\mu}\right)^* = \frac{e^4}{(s-m^2)(u-m^2)}\left(8m^2s + 8m^2u + 16m^4\right)$$
$$\mathcal{M}_{2ab}{}^{\nu\mu}\left(g_{\nu\alpha}\mathcal{M}_{2ab}{}^{\alpha\beta}g_{\beta\mu}\right)^* = \frac{e^4}{(u-m^2)^2}\left(-8su + 8m^2s + 24m^2u + 8m^4\right)$$

where s, t, and u are Mandelstam variables

$$s = (p_1 + p_2)^2$$
$$t = (p_1 - p_3)^2$$
$$u = (p_1 - p_4)^2$$

For clarity define

$$f_{11} = -8su + 24m^2s + 8m^2u + 8m^4$$
$$f_{12} = 8m^2s + 8m^2u + 16m^4$$
$$f_{22} = -8su + 8m^2s + 24m^2u + 8m^4$$

and write $\langle|\mathcal{M}|^2\rangle$ more compactly as

$$\langle|\mathcal{M}|^2\rangle = \frac{e^4}{4}\left(\frac{f_{11}}{(s-m^2)^2} + \frac{2f_{12}}{(s-m^2)(u-m^2)} + \frac{f_{22}}{(u-m^2)^2}\right)$$

For $E \gg m$ a useful approximation is to set $m = 0$ and obtain

$$\begin{aligned} f_{11} &= -8su \\ f_{12} &= 0 \\ f_{22} &= -8su \end{aligned}$$

Then $\langle|\mathcal{M}|^2\rangle$ simplifies as

$$\langle|\mathcal{M}|^2\rangle = \frac{e^4}{4}\left(-\frac{8su}{s^2} - \frac{8su}{u^2}\right) = -2e^4\left(\frac{u}{s} + \frac{s}{u}\right)$$

For $m = 0$ the Mandelstam variables are

$$\begin{aligned} s &= 4E^2 \\ t &= -2E^2(1 - \cos\theta) \\ u &= -2E^2(1 + \cos\theta) \end{aligned}$$

Hence

$$\langle|\mathcal{M}|^2\rangle = 2e^4\left(\frac{1 + \cos\theta}{2} + \frac{2}{1 + \cos\theta}\right)$$

The differential cross section is

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{e^4}\frac{\langle|\mathcal{M}|^2\rangle}{4s} = \frac{\alpha^2}{8E^2}\left(\frac{1 + \cos\theta}{2} + \frac{2}{1 + \cos\theta}\right)$$

Multiply by $(\hbar c)^2$ to convert from natural units to physical units.

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2}{8E^2}\left(\frac{1 + \cos\theta}{2} + \frac{2}{1 + \cos\theta}\right)(\hbar c)^2$$

We also have

$$\frac{d\sigma}{d\cos\theta} = 2\pi\frac{d\sigma}{d\Omega} = \frac{\pi\alpha^2}{s}\left(\frac{\cos\theta + 1}{2} + \frac{2}{\cos\theta + 1}\right)(\hbar c)^2$$

See equation (1) of arxiv.org/pdf/hep-ex/0504012.

Eigenmath script